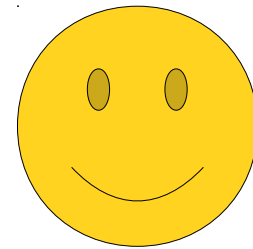
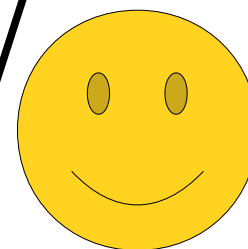


作业 0：使用调试器

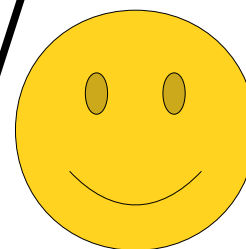
大家好！



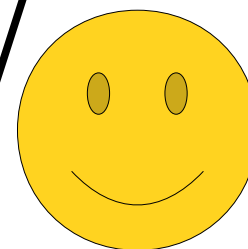
在作业 0 中，我们希望你先练习一下如何在 Qt Creator
中使用调试器。



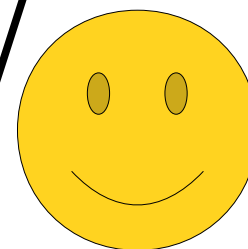
调试器是一种工具，可以帮助你在程序运行时查看它正在做什么。



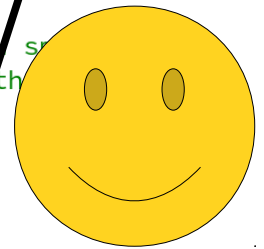
它非常适合用来查找程序中的错误。练习得越多，你就越容易修正自己编写的程序中的问题。



你可以把这份指南看作一个简短的操作教程：它会帮助你了解如何使用调试器，以及如何理解眼前看到的信息。



首先，打开你在本次作业第一部分运行过的 Name Hash 程序。向下滚动到 nameHash 函数，让整个函数都显示在窗口中。



```
42 * For t
43 * treat
44 * It th
45 * F_p, wh
46 * some smaller prime number q (you didn't expect a
47 * but we thought it might be fun!)
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers, a large prime p and a small
51     * prime. These numbers were chosen because their product is less than 2^31.
52     * 2^31 - kLargePrime - 1.
53     */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first name, then the last
60     * name, updating the hash at each step.
61     */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. The numeric values of
64         * lower-case letters are always less than 127.
65         */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Activities Qt Creator Jan 4 3:09 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

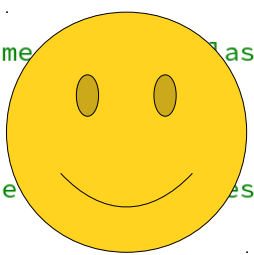
Projects NameHash [main] NameHash.pro Sources NameHash.cpp

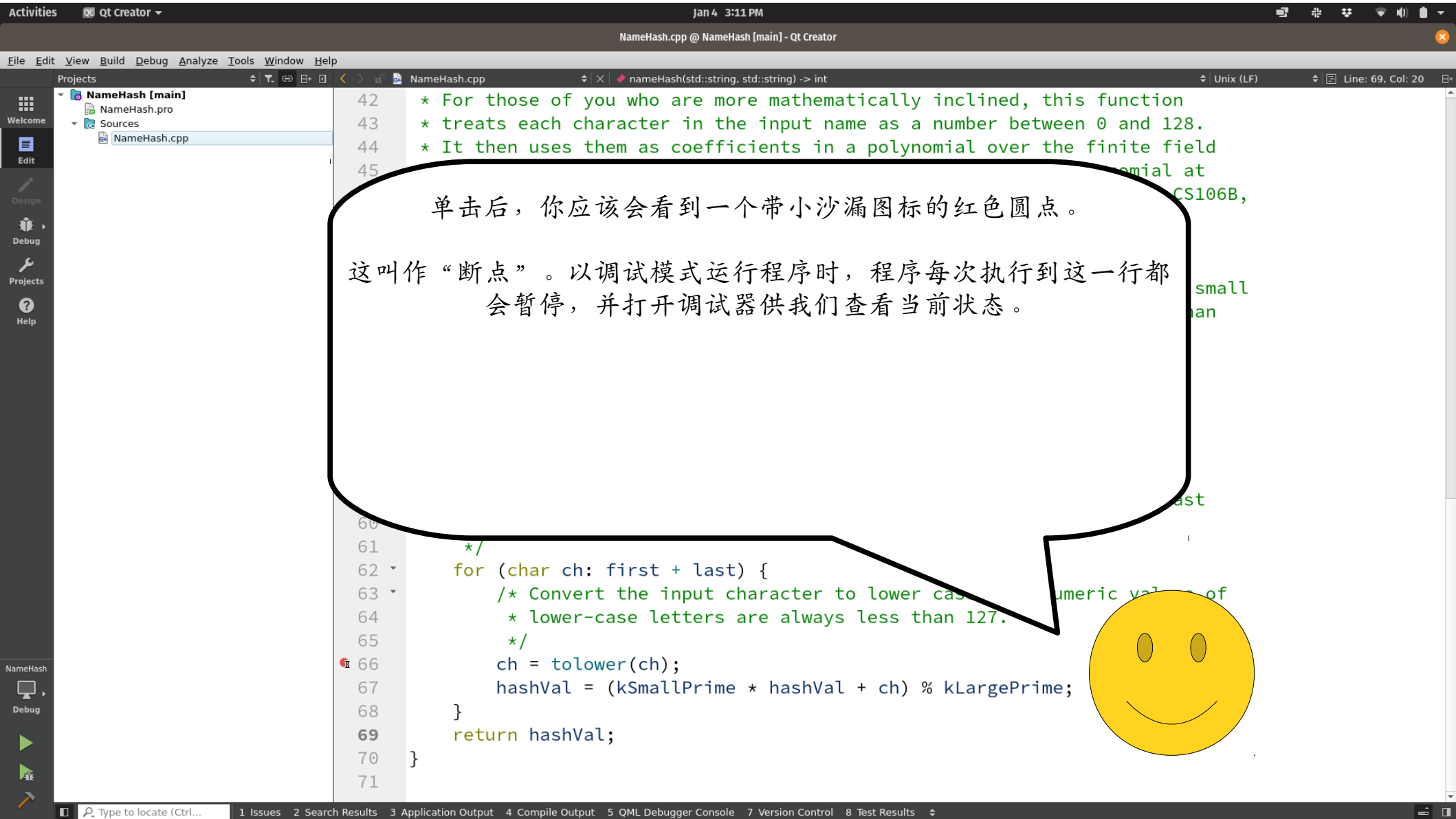
42 * For those of you who are more mathematically inclined, this function
43 * treats each character in the input name as a number between 0 and 128.
44 * It then uses them as coefficients in a polynomial over the finite field
45 * F_p , where p is a large prime number, and evaluates that polynomial at $x = \text{first} + \text{last}$ for CS106B,

把鼠标移到第 66 行行号左侧的空白区域，然后单击鼠标！

56
57 `int hashVal = 0;`
58
59 `/* Iterate across all the characters in the input name, from first to last`
60 `* name, updating the hash at each step.`
61 `*/`
62 `for (char ch: first + last) {`
63 `/* Convert the input character to lower case. The value of`
64 `* lower-case letters are always less than 127.`
65 `*/`
66 `ch = tolower(ch);`
67 `hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;`
68 `}`
69 `return hashVal;`
70 `}`
71

and a small
less than





Activities Qt Creator Jan 4 3:11 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

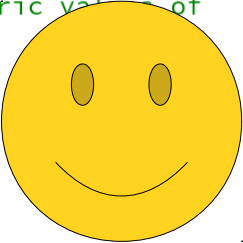
```
42 * For those of you who are more mathematically inclined, this function
43 * treats each character in the input name as a number between 0 and 128.
44 * It then uses them as coefficients in a polynomial over the finite field
45 * F_p, where p is a large prime number, and evaluates that polynomial at
46 * some smaller prime number q. (You aren't expected to know this for CS106B,
47 * but we thought it might be fun!)
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers, a large prime and a small
51     * prime, such that the large prime is a multiple of the small prime plus
52     * one. (You aren't expected to know this for CS106B, but we thought it
53     * might be fun!) */
54     const int kLargePrime = 1000000007;
55     const int kSmallPrime = 1000000003;
56     int hashVal = 0;
57     for (char ch: first) {
58         /* Convert the input character to lower case. The numeric value of
59         * lower-case letters are always less than 127. */
60         ch = tolower(ch);
61         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
62     }
63     for (char ch: last) {
64         /* Convert the input character to lower case. The numeric value of
65         * lower-case letters are always less than 127. */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

现在我们要以调试模式运行程序。请单击屏幕左下角的“以调试模式运行”按钮，它就在普通绿色“运行”按钮的下方。单击之后…

...

at

numeric value of



……你应该会看到类似这样的界面！注意，Qt Creator 中多出了不少面板。稍后我们会说明这些面板分别表示什么。

The screenshot shows the Qt Creator IDE interface for a project named 'NameHash'. The main editor displays the following C++ code:

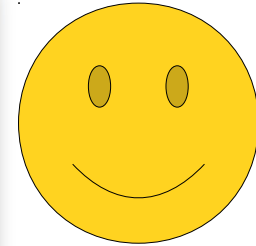
```
45  
46 * some smaller prime number q. (You aren  
47 * but we thought it might be fun!)  
48 */  
49  
50  
51  
52  
53  
54  
55  
56  
57  
58  
59  
60  
61  
62  
63  
64  
65  
66 ch = tolower(ch);  
67 hashVal = (kSmallPrime * hashVal + ch) % kl  
68 }  
69 return hashVal;
```

A 'NameHash Console' window is open, displaying the prompt 'What is your first name?'. A yellow smiley face is drawn next to the console. The bottom status bar shows the debugger is active, and the application has started.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1		...g	...eHash.cpp	66	...	...	...	5555b6782				(all)

与此同时，按图示输入名字 Ada，然后按回车。这里请务必输入 Ada，而不是你自己的名字。

(当然，除非你就叫 Ada。)



Activities NameHash

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Welcome Edit Design Debug Projects Help

```
45
46 * some smaller prime number q. (You aren't
47 * but we thought it might be fun!)
48 */
49
50
51
52 What is your first name? Ada
53 What is your last name?
54
55
56
57
58
59
60
61
62
63
64
65
66 ch = tolower(ch);
67 hashVal = (kSmallPrime * hashVal + ch) % kl
68 }
69 return hashVal;
```

NameHash Console

File Edit Options Help

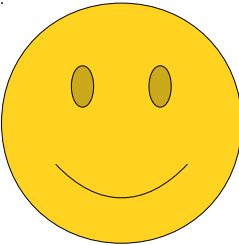
What is your first name? Ada
What is your last name?

Debugger GDB for "NameHash" Threads: #12 Application started.

Level	Function	File	Line	Address	Number	Funct File	Line	Address	Condition	Ignore	Threads
1		...g) ...eHash.cpp	66	...5555b6782							(all)

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

现在输入姓氏 Lovelace，但先不要按回车！

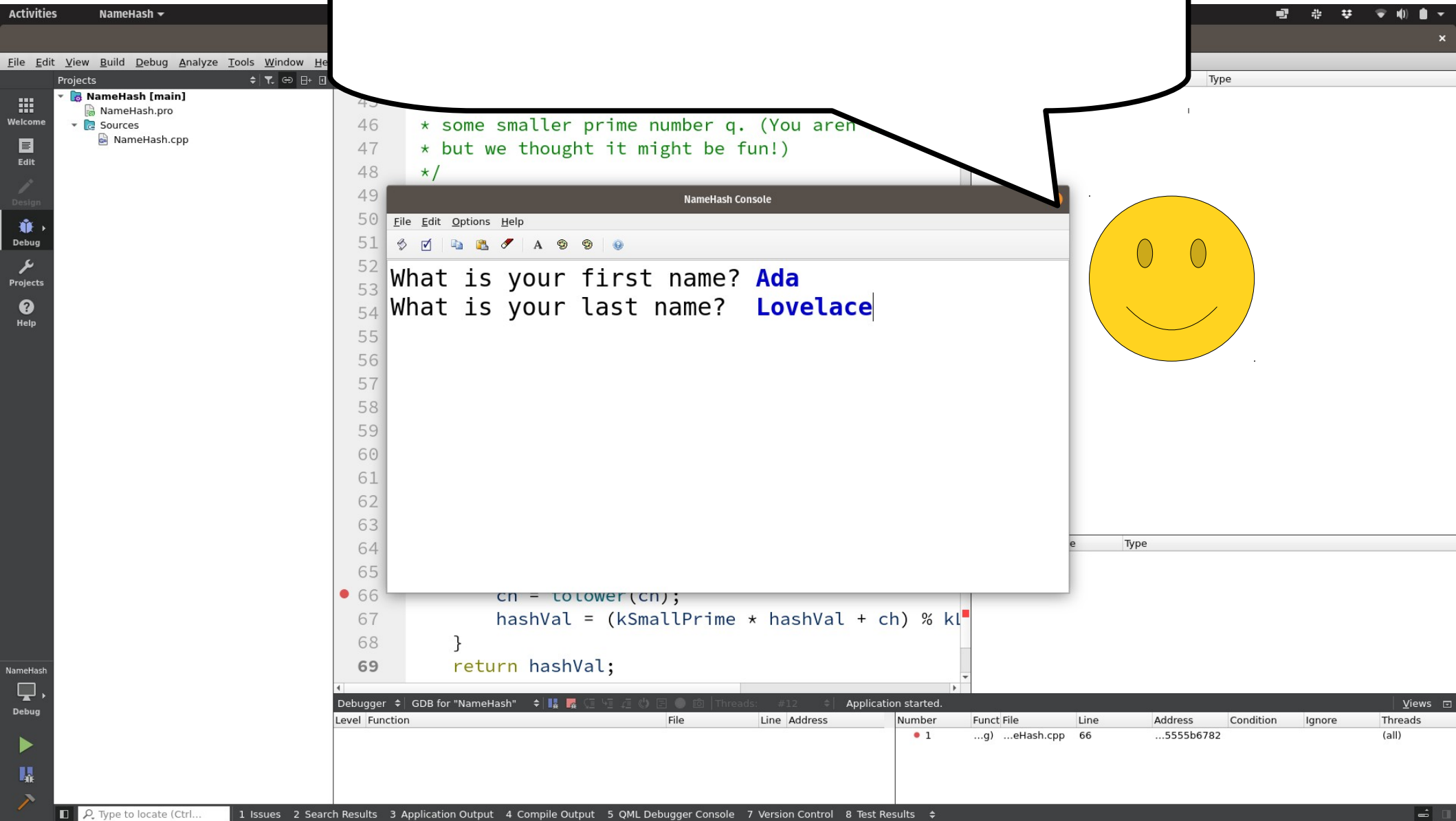


```

46 * some smaller prime number q. (You aren't
47 * but we thought it might be fun!)
48 */
49
50
51
52 What is your first name? Ada
53 What is your last name? Lovelace
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69

```

按下回车后，Qt Creator
中会弹出很多内容。不要慌，这是正常现象。



The screenshot shows the Qt Creator IDE interface. The left sidebar contains the 'Projects' panel with 'NameHash [main]' selected, showing 'NameHash.pro' and 'NameHash.cpp'. The main editor displays a C++ file with a hash function. The 'NameHash Console' window is open, showing the program's output: 'What is your first name? Ada' and 'What is your last name? Lovelace'. A yellow smiley face is drawn next to the console. The background code shows a hash function implementation. The bottom status bar shows the debugger is running, with 'Application started.' and a table of threads.

```
45  
46 * some smaller prime number q. (You aren  
47 * but we thought it might be fun!)  
48 */  
49  
50  
51  
52 What is your first name? Ada  
53 What is your last name? Lovelace  
54  
55  
56  
57  
58  
59  
60  
61  
62  
63  
64  
65  
66 ch = tolower(ch);  
67 hashVal = (kSmallPrime * hashVal + ch) % kl  
68 }  
69 return hashVal;
```

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1		...g)	...eHash.cpp	66	...	...	...	5555b6782				(all)

现在按下回车，看看接下来会发生什么

!

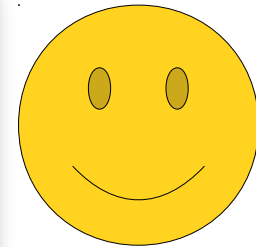
```
45  
46 * some smaller prime number q. (You aren  
47 * but we thought it might be fun!)  
48 */  
49
```

```
50  
51  
52 What is your first name? Ada  
53 What is your last name? Lovelace  
54  
55  
56  
57  
58  
59  
60  
61  
62  
63  
64  
65
```

```
66     ch = tolower(ch);  
67     hashVal = (kSmallPrime * hashVal + ch) % kl  
68 }  
69 return hashVal;
```

Debugger GDB for "NameHash" Threads: #12 Application started.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
					1	...	...eHash.cpp	66	...5555b6782			(all)



看！我们回到了 Qt Creator，界面各处出现了大量数值。



```

48
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers
51      * prime. These numbers were chosen because the
52      *  $2^{31} - \text{kLargePrime} - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71

```

Name	Value	Type
last	16908799	std::string
	127	int
	0	int
	"Ada"	std::string
	'A'	char
	65	0x41
	"AdaLovelace"	std::string
	16908799	std::string::iterator
	127	std::string::iterator
	0	std::string::iterator



Debugger ▾ GDB for "NameHash" ▾ [Icons] Threads: #12 NameHash ▾ Stopped at breakpoint 1 in thread 12. ▾ Views ▾												
Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::Function_handler<int (), QtGui::startBackgroundEve...			0x555555b616bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Build

这里的信息很多。我们来看看究竟发生了什么。

```
48
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers
51     * prime. These numbers were chosen because they are
52     *  $2^{31} - kLargePrime - 1$ .
53     */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first and last
60     * name, updating the hash at each step.
61     */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case. Note that
64         * lower-case letters are always less than upper-case
65         */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

```
"AdaLovelace"
'A' 65
"Ada"
0
16908799
127
"Lovelace"
```



Type
std::string::iterator
std::string::iterator
std::string &&
char
std::string
int
int
int
std::string

Name	Value	Type
------	-------	------

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Build

Activities Qt Creator Jan 4 3:15 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - \text{kLargePrime} - 1$ .
```

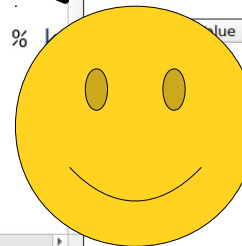
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x5555555b6782
2	studentMain	NameHash.cpp	31	0x5555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Build

首先，请注意：红色断点图标中现在出现了一个黄色箭头。



Activities Qt Creator Jan 4 3:15 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers;
51        * prime. These numbers were chosen because the
52        *  $2^{31} - \text{kLargePrime} - 1$ .
```

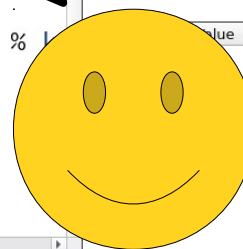
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x5555555b6782
2	studentMain	NameHash.cpp	31	0x5555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
4	GThreadStd::run()			0x55555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Build

这个黄色箭头表示程序当前执行到的位置。程序之所以停在这一行，是因为触发了你刚才设置的断点。



Activities Qt Creator Jan 4 3:15 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers;
51        * prime. These numbers were chosen because the
52        *  $2^{31} - \text{kLargePrime} - 1$ .
```

62 for (char ch: first + last)
63 /* Convert the input character to lower-case
64 * lower-case letters are always
65 */
66 ch = tolower(ch);
67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68 }
69 return hashVal;
70 }
71 }

66

每次打开调试器时，首先都应该确认程序当前停在哪里。因此，请养成寻找这个黄色箭头的习惯。

Value Type

__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	"A" 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	g) ...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Build

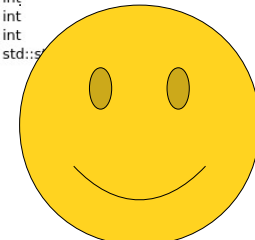
Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string



接下来看看这个面板。它叫作“调用栈”。

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

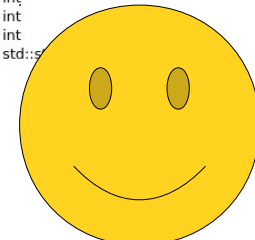
Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::s



现在我们知道自己位于 nameHash 函数中，因为黄色箭头清楚地指出了当前所在的代码行。

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

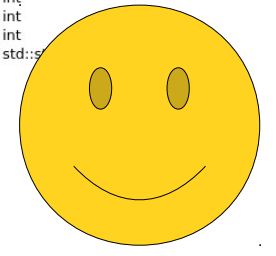
Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::s



不过，黄色箭头无法告诉我们程序是怎样运行到这里的。究竟是哪一部分代码调用了 nameHash？

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65 0x41	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::s

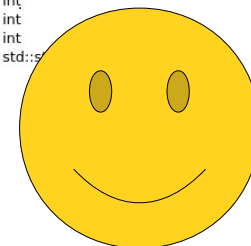


调用栈正好可以告诉我们答案!

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	g) ...eHash.cpp	66	...	5555b6782		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread create.c	463	0x7ffff6257590								

```
48  */
49  int nameHash(string first, string last){
50  /* This hashing scheme needs two prime numbers:
51   * prime. These numbers were chosen because they are
52   *  $2^{31} - kLargePrime - 1$ .
53   */
54  static const int kLargePrime = 16908799;
55  static const int kSmallPrime = 127;
56
57  int hashVal = 0;
58
59  /* Iterate across all the characters in the first
60   * name, updating the hash at each step.
61   */
62  for (char ch : first)
63  {
64      /*
65       *
66       */
67      hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68  }
69  return hashVal;
70 }
71
```

Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	"A" 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string



注意，调用栈会按顺序列出一系列函数。最上方是 nameHash（当前所在位置），它的下一层是 studentMain。

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Build

Activities Qt Creator Jan 4 3:15 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

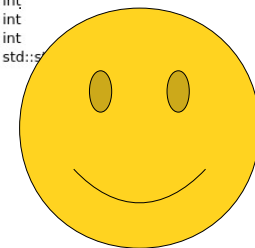
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch : first)
63     {
64         /*
65          *
66          */
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	NameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...	...		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::call_once(handler_list(), QtGui::testBackendEnv::init())			0x5555555161b								
4	GThreadSt...			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start threa...	pthread_create.c	463	0x7ffff6257590								

Build

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



双击调用栈第 2 层中的 studentMain。然后……

Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVa
34     return 0;
35 }
36
37 /* This is the actual function that c
38 * to talk more about what hash functi
39 * the meantime, think of it as a func
40 * of the input and produces a number
41 *
42 * For those of you who are more mathem
43 * treats each character in the input name as a num
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funcnt	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	g) ...eHash.cpp	66	...	55555b6782		(all)
2	studentMain	NameHash.cpp	31	0x5555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVa
34     return 0;
35 }
36
37 /* This is the actual function that c
38 * to talk more about what hash funct
39 * the meantime, think of it as a fun
40 * of the input and produces a number
41 *
42 * For those of you who are more math
43 * treats each character in the input
```

Debugger GDB for "NameHash" Threads: #12 Na

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x55
2	studentMain	NameHash.cpp	31	0x55
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55
4	GThreadStd::run()			0x55
5	??			0x7ff
6	start thread	pthread create.c	463	0x7ff

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Con

Views Threads (all)

注意，黄色箭头指向第 31 行。该行调用了 nameHash 函数。正是这部分代码调用了 nameHash，因此程序才会运行到设置断点的那一行。

Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

21 #include "simpio.h" // for getLine

22 using namespace std;

23 /* Prototype for the nameHash function. This lets u

24 * in main and then define it later in the program.

25 */

26 int nameHash(string first, string last);

27 int main() {

28 string first = getLine("What is your first name

29 string last = getLine("What is your last name?

30

31 int hashValue = nameHash(first, last);

32

33 cout << "The hash of your name is: " << hashVa

34 return 0;

35 }

36

37 /* This is th

38 * to talk mo

39 * the meanti

40 * of the inp

41 *

42 * For those

43 * treats each ch

Name

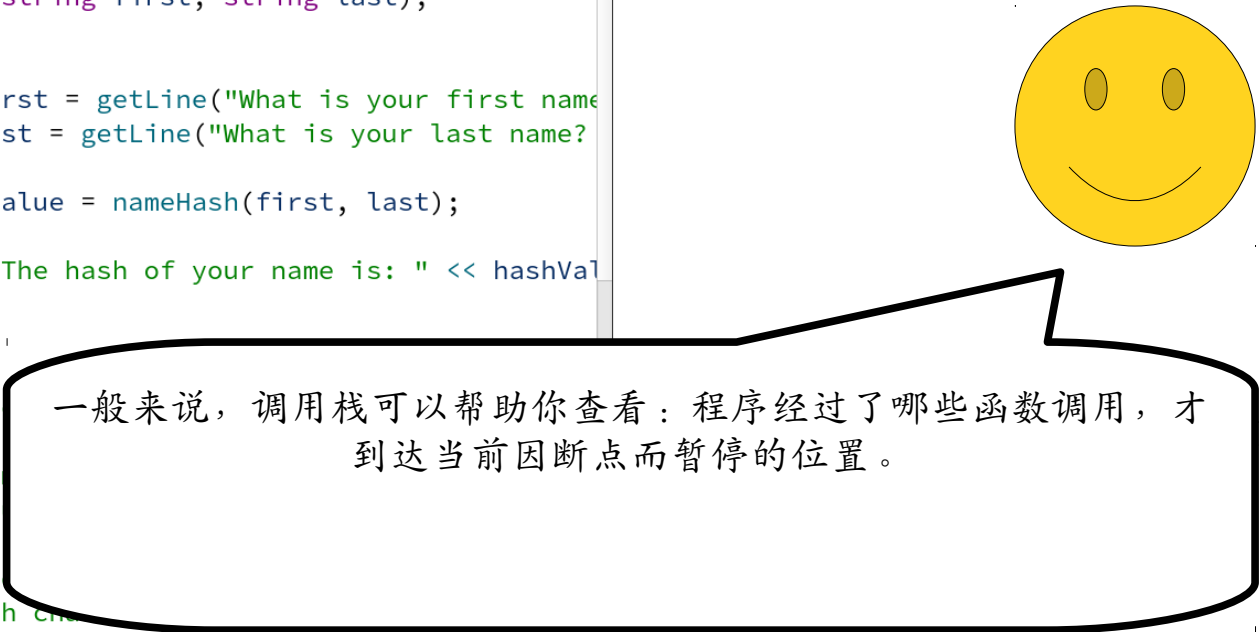
Name	Value	Type
first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782								
2	studentMain	NameHash.cpp	31	0x555555b6595	1	...	...eHash.cpp	66	...5555b6782			(all)
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

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1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout <<
34     return
35 }
36
37 /* This is
38 * to talk
39 * the meant
40 * of the input and produces a number.
41 *
42 * For those of you who are more mathematically inc
43 * treats each character in the input name as a num
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string



根据操作系统的不同，你可能会在 studentMain 下方看到一些额外的函数。它们是什么？

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782								
2	studentMain	NameHash.cpp	31	0x555555b6595	1	...	...eHash.cpp	66	...5555b6782			(all)
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

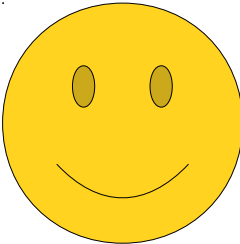
File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout <<
34     return
35 }
36
37 /* This is
38 * to talk
39 * the meant
40 * of the input and produces a number.
41 *
42 * For those of you who are more mathematically inc
43 * treats each character in the input name as a num
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string



这些灰色函数是辅助函数；我们的库会自动调用它们，以完成程序启动前的准备工作。

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	g) ...eHash.cpp	66	...	5555b6782		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread create.c	463	0x7ffff6257590								

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

21 #include "simpio.h" // for getLine

22 using namespace std;

23 /* Prototype for the nameHash function. This lets u

24 * in main and then define it later in the program.

25 */

26 int nameHash(string first, string last);

27 int main() {

28 string first = getLine("What is your first name

29 string last = getLine("What is your last name?

30

31 int hashCode = nameHash(first, last);

32

33 cout <<

34 return

35 }

36

37 /* This is

38 * to talk

39 * the meant

40 * of the input and produces a number.

41 *

42 * For those of you who are more mathematically inc

43 * treats each character in the input name as a num

Name

Name	Value	Type
first	"Ada"	std::string
hashCode	0	int
last	"Lovelace"	std::string

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

不用担心这些函数。它们会出现在你运行的各个程序中，可以直接忽略。



Activities Qt Creator Jan 4 3:22 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout <<
34     return
35 }
36
37 /* This is
38 * to talk
39 * the meant
40 * of the input and produces a number.
41 *
42 * For those of you who are more mathematically inc
43 * treats each character in the input name as a num
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

现在回到 nameHash 函数。双击调用栈顶部的 nameHash 条目。然后……

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	std::function<int ()> QtGui::startBackgroundEve...	libQtGui.so.5	3	0x555555b6782								
3	std::function<int ()> QtGui::startBackgroundEve...			0x555555b616c								
4	GThreadSt			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

你会立刻回到这里!



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

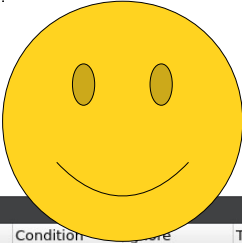
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* I
60      * r
61      */
62     for
63
64
65
66     ch = tolower(ch);
67     hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68 }
69 return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
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4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

快速回顾一下目前看到的内容。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

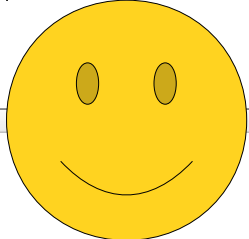
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54
55
56
57
58
59
60
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

要设置断点并让程序在某处暂停，请单击目标行号左侧的边栏。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

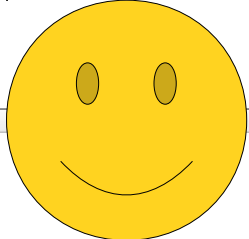
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they
52      * are 2^31 - kLargePrime - 1.
53      */
54
55
56
57
58
59
60
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
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70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

程序到达断点后，调试器会显示各种有用的信息。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

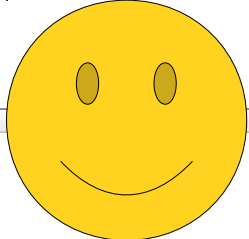
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54
55
56
57
58
59
60
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

黄色箭头指出程序当前所在的位置。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

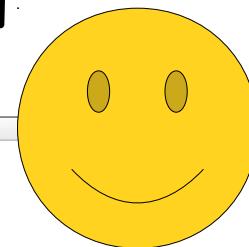
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54
55
56
57
58
59
60
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), Qt...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

调用栈显示程序是如何进入当前函数的。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - \text{kLargePrime} - 1$ .
53      */
54
55
56
57
58
59
60
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

现在看看如何读取这个函数中各变量的值。

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	"A" 65 0x41	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

请看右上方的这个面板。



The image shows the Qt Creator IDE interface. The main editor displays the `NameHash.cpp` file. The code defines a `nameHash` function that takes two strings, `first` and `last`, and returns a hash value. The function uses two prime numbers, `kLargePrime` (16908799) and `kSmallPrime` (127), to calculate the hash. The code is as follows:

```
nameHash(string first, string last){
    /* This hashing scheme needs two prime numbers:
    * prime. These numbers were chosen because the
    * 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

The right sidebar shows the Variable Explorer, which displays the variables and their values for the current scope. The variables are:

Name	Value	Type
<code>__for_begin</code>	<code>@0x7fffc6058c78</code>	<code>std::string::iterator</code>
<code>__for_end</code>	<code>@0x7fffc6058c80</code>	<code>std::string::iterator</code>
<code>__for_range</code>	<code>"AdaLovelace"</code>	<code>std::string &&</code>
<code>ch</code>	<code>'A'</code> 65	<code>char</code>
<code>first</code>	<code>"Ada"</code>	<code>std::string</code>
<code>hashVal</code>	<code>0</code>	<code>int</code>
<code>kLargePrime</code>	<code>16908799</code>	<code>int</code>
<code>kSmallPrime</code>	<code>127</code>	<code>int</code>
<code>last</code>	<code>"Lovelace"</code>	<code>std::string</code>

The bottom status bar shows the debugger is stopped at breakpoint 1 in thread 12. The stack trace is as follows:

Level	Function	File	Line	Address
1	<code>nameHash</code>	<code>NameHash.cpp</code>	66	<code>0x555555b6782</code>
2	<code>studentMain</code>	<code>NameHash.cpp</code>	31	<code>0x555555b6595</code>
3	<code>std::_Function_handler<int (), QtGui::startBackgroundEve...</code>			<code>0x5555556161bc</code>
4	<code>GThreadStd::run()</code>			<code>0x555555f9476</code>
5	<code>??</code>			<code>0x7ffff6143d84</code>
6	<code>start_thread</code>	<code>pthread_create.c</code>	463	<code>0x7ffff6257590</code>

这个窗口会显示当前作用域内所有局部变量的值。



The image shows the Qt Creator IDE interface with a C++ project named "NameHash". The main editor displays the source code for "NameHash.cpp". A red dashed box highlights the "Variables" window on the right, which lists the current scope's variables. The "Debugger" window at the bottom shows the execution state, including a call stack and a list of threads.

```
1  NameHash (main)
2  NameHash.pro
3  Sources
4  NameHash.cpp

NameHash.cpp
1  //
2  // This hashing scheme needs two prime numbers:
3  // prime. These numbers were chosen because they
4  // are 2^31 - kLargePrime - 1.
5  //
6  static const int kLargePrime = 16908799;
7  static const int kSmallPrime = 127;
8
9  int hashVal = 0;
10
11 /* Iterate across all the characters in the first
12 * name, updating the hash at each step.
13 */
14 for (char ch: first + last) {
15     /* Convert the input character to lower case
16     * lower-case letters are always less than
17     */
18     ch = tolower(ch);
19     hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
20 }
21 return hashVal;
22 }
```

Variables window:

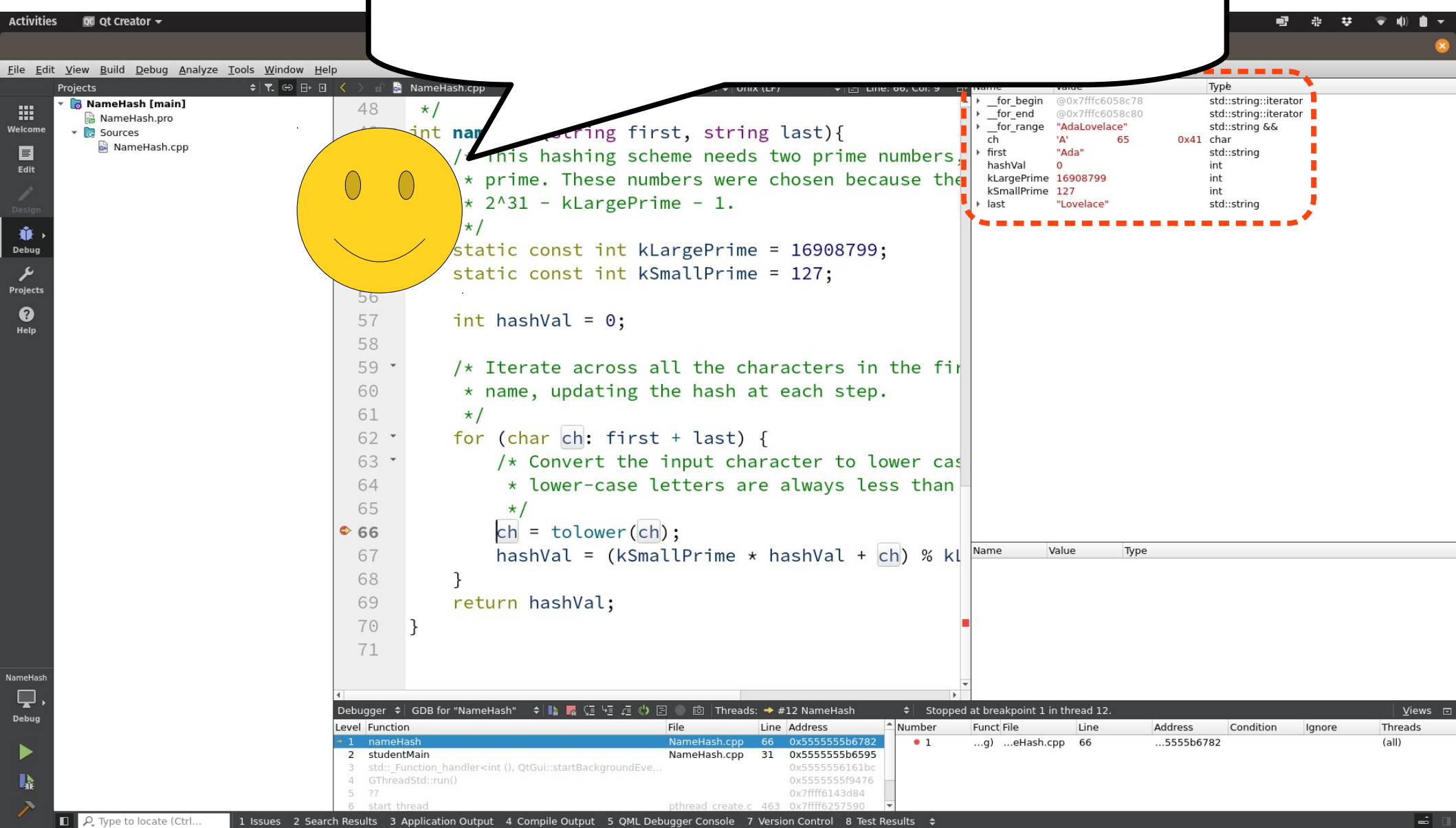
Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65 0x41	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger window:

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Number	Func	File	Line	Address	Condition	Ignore	Threads
1	...	...eHash.cpp	66	...	...		(all)

根据操作系统的不同，这些变量的排列顺序可能不同；除了
ch、hashVal
这类熟悉的变量外，还可能出现一些看起来很奇怪的条目。



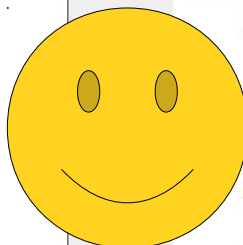
The image shows the Qt Creator IDE with the NameHash.cpp file open. The code defines a hash function for a string. A yellow smiley face is placed over the code. A red dashed box highlights the variable viewer on the right, which shows the state of variables at the current execution point. The variable viewer lists variables such as `__for_begin`, `__for_end`, `__for_range`, `ch`, `first`, `hashVal`, `kLargePrime`, `kSmallPrime`, and `last`. The bottom panel shows the debugger's call stack, indicating the current function is `nameHash` at line 66.

```
48  */
49  int nameHash(string first, string last){
50      /* This hashing scheme needs two prime numbers:
51       * prime. These numbers were chosen because they
52       * are 2^31 - kLargePrime - 1.
53       */
54      static const int kLargePrime = 16908799;
55      static const int kSmallPrime = 127;
56
57      int hashVal = 0;
58
59      /* Iterate across all the characters in the first
60       * name, updating the hash at each step.
61       */
62      for (char ch: first + last) {
63          /* Convert the input character to lower case
64           * lower-case letters are always less than
65           */
66          ch = tolower(ch);
67          hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68      }
69      return hashVal;
70  }
71  }
```

Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	"A"	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

忽略那些奇怪的条目后，我们就能看到一些熟悉的变量名。



Qt Creator interface showing the NameHash.cpp file and the GDB debugger.

Code Snippet (NameHash.cpp):

```
48 */
49 int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers:
    * prime. These numbers were chosen because they
    * are 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

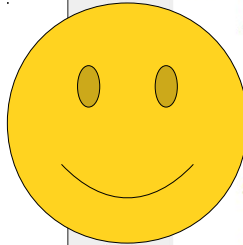
Debugger Variables:

Name	Value	Type
__for_end	@0x7fff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A' 65	std::string && char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger Console:

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

例如，这里可以看到 kLargePrime 和 kSmallPrime 的值，它们与声明时指定的值一致。



```
48  */
49  int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they are
    *  $2^{31} - kLargePrime - 1$ .
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

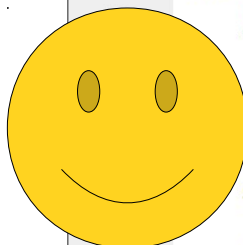
    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Name	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger ▾ GDB for "NameHash" ▾					Threads: ➔ #12 NameHash		Stopped at breakpoint 1 in thread 12.					Views ▾
Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
➔ 1	nameHash	NameHash.cpp	66	0x555555b6782	● 1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

还可以看到，此时 hashVal 仍然是 0。



```
48  */
49  int nameHash(const string& first, string& last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they
    * are 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

```
__for_end @0x7ffff6058c80
__for_range "AdaLovelace"
ch 'A' 65
first "Ada"
hashVal 0
kLargePrime 16908799
kSmallPrime 127
last "Lovelace"
```

```
type
std::string::iterator
std::string::iterator
std::string &&
char
std::string
int
int
int
std::string
```

Name	Value	Type
------	-------	------

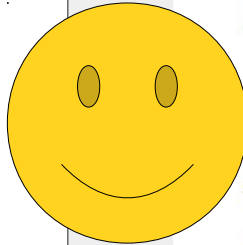
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1		...g) ...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

接下来逐行执行程序时，我们会看到这些值发生变化。



Qt Creator interface showing the NameHash.cpp file and the GDB debugger.

Code Editor (NameHash.cpp):

```
48  */
49  int nameHash(const string& first, string& last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they are
    *  $2^{31} - kLargePrime - 1$ .
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first string,
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than 128.
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Debugger (GDB for "NameHash"):

Threads: #12 NameHash

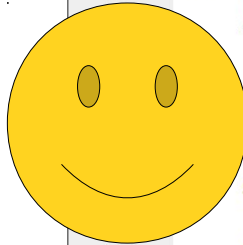
Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Variable Watcher:

Name	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

现在看看这个 for 循环。



```
48  */
49  int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers:
    * prime. These numbers were chosen because they are
    *  $2^{31} - \text{kLargePrime} - 1$ .
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first string
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than 128.
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

```
__for_end @0x7ffff6058c80
__for_range "AdaLovelace"
ch 'A' 65 0x41
first "Ada"
hashVal 0
kLargePrime 16908799
kSmallPrime 127
last "Lovelace"
```

```
type
std::string::iterator
std::string::iterator
std::string &&
char
std::string
int
int
int
std::string
```

Name	Value	Type
------	-------	------

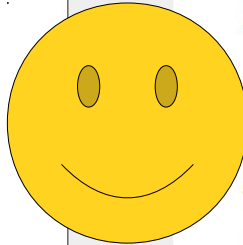
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

这是一个基于范围的 for 循环。它表示：“依次取出字符串 first + last 中的每个字符，并对该字符执行相应操作。”



```
48  */
49  int nameHash(const string& first, string& last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they are
    *  $2^{31} - kLargePrime - 1$ .
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first string
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than 128.
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Variable	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

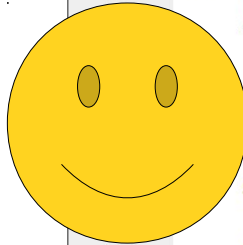
Name	Value	Type
------	-------	------

Debugger: GDB for "NameHash" | Threads: #12 NameHash | Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl+L) | 1 Issues | 2 Search Results | 3 Application Output | 4 Compile Output | 5 QML Debugger Console | 7 Version Control | 8 Test Results

还记得之前输入的名字 Ada Lovelace 吗?



```
48 */
49 int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers:
    * prime. These numbers were chosen because they
    * are 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Variable	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

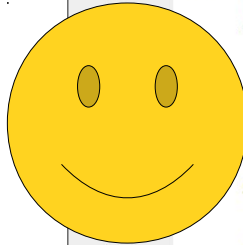
Name	Value	Type
------	-------	------

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

查看变量 ch 的当前值，可以看到它是 A，也就是 Ada Lovelace 的第一个字母。



```
48  */
49  int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they
    * are 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

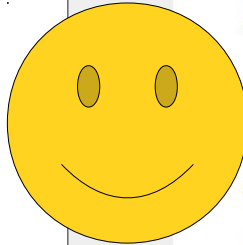
    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Name	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"Ada Lovelace"	std::string::iterator
ch	'A'	char
first	"Ada Lovelace"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger					Threads				
GDB for "NameHash"					#12 NameHash				
Level	Function	File	Line	Address	Number	Funct	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	g) ...eHash.cpp	66	...5555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595					
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc					
4	GThreadStd::run()			0x555555f9476					
5	??			0x7ffff6143d84					
6	start thread	pthread_create.c	463	0x7ffff6257590					

现在我们已经知道：程序停在第 66 行；main 调用了 nameHash；并且可以看到此时程序中各变量的值。



```
48  */
49  int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they
    * are 2^31 - kLargePrime - 1.
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

		type	
__for_end	@0x7ffff6058c80	std::string::iterator	
__for_range	"AdaLovelace"	std::string::iterator	
ch	'A'	std::string &&	
first	"Ada"	char	0x41
hashVal	0	std::string	
kLargePrime	16908799	int	
kSmallPrime	127	int	
last	"Lovelace"	std::string	

Name	Value	Type
------	-------	------

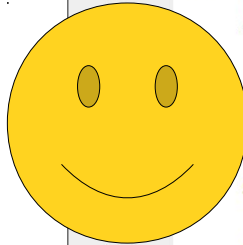
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)
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3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

接下来做一件很有意思的事：让程序每次只执行一行，并观察每一步发生了什么！



```
48  */
49  int nameHash(const string& first, string last){
    /* This hashing scheme needs two prime numbers.
    * prime. These numbers were chosen because they are
    *  $2^{31} - \text{kLargePrime} - 1$ .
    */
    static const int kLargePrime = 16908799;
    static const int kSmallPrime = 127;

    int hashVal = 0;

    /* Iterate across all the characters in the first string
    * name, updating the hash at each step.
    */
    for (char ch: first + last) {
        /* Convert the input character to lower case.
        * lower-case letters are always less than 128.
        */
        ch = tolower(ch);
        hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
    }
    return hashVal;
}
```

Variable	Value	Type
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string::iterator
ch	'A'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Debugger										Threads			
GDB for "NameHash"										#12 NameHash			
										Stopped at breakpoint 1 in thread 12.			
Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads	
1	nameHash	NameHash.cpp	66	0x555555b6782	1	...	...eHash.cpp	66	...5555b6782			(all)	
2	studentMain	NameHash.cpp	31	0x555555b6595									
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc									
4	GThreadStd::run()			0x555555f9476									
5	??			0x7ffff6143d84									
6	start_thread	pthread_create.c	463	0x7ffff6257590									

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator

Jan 4 3:30 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

NameHash [main]

- NameHash.pro
- Sources
 - NameHash.cpp

48 `*/`

49 `int nameHash(string first, string last){`

50 `/* This hashing scheme needs two prime numbers;`

51 `* prime. These numbers were chosen because the`

52 `*  $2^{31} - kLargePrime - 1$ .`

53 `*/`

54 `static const int kLargePrime = 16908799;`

55 `static const int kSmallPrime = 127;`

56

57 `int hashVal = 0;`

58

59 `/* Iterate across all the characters in the fir`

60 `* name, updating the hash at each step.`

61 `*/`

62 `for (ch`

63 `/*`

64 `*`

65 `*/`

66 `ch`

67 `has`

68 `}`

69 `return hashVal;`

70 `}`

71

Debugger GDB for "NameHash"

Threads: #12 NameHash

Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

在调用栈上方，可以看到几个小按钮图标。



Activities Qt Creator Jan 4 3:30 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch : first)
63     {
64         /*
65          *
66          */
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

这些按钮可以让你继续程序、停止程序、逐行执行程序等。



Activities Qt Creator

Jan 4 3:30 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers:

51 * prime. These numbers were chosen because the

52 * $2^{31} - kLargePrime - 1$.

53 */

54 static const int kLargePrime = 16908799;

55 static const int kSmallPrime = 127;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the fir

60 * name, updating the hash at each step.

61 */

62 for (ch

63 /*

64 *

65 */

66 ch

67 has

68 }

69 return hashVal;

70 }

71 }

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start thread	pthread_create.c	463	0x7ffff6257590

Debugger Variables

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65 0x41	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

把鼠标悬停在左起第三个按钮上。悬停后，提示文字应为“单步跳过 (Step Over)”。



Activities Qt Creator

Jan 4 3:30 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

```
48  */
49  int nameHash(string first, string last){
50  /* This hashing scheme needs two prime numbers:
51   * prime. These numbers were chosen because the
52   *  $2^{31} - kLargePrime - 1$ .
53   */
54  static const int kLargePrime = 16908799;
55  static const int kSmallPrime = 127;
56
57  int hashVal = 0;
58
59  /* Iterate across all the characters in the first
60   * name, updating the hash at each step.
61   */
62  for (char ch : first)
63  {
64      /*
65       *
66       */
67      hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68  }
69  return hashVal;
70 }
71
```

Debugger

GDB for "NameHash"

Threads: #12 NameHash

Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x5555555b6782
2	studentMain	NameHash.cpp	31	0x5555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundThread::...>::operator()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Debugger Console

Type to locate (Ctrl+K)

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Debugger Variables

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'A' 65	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

确认你选择的是“单步跳过 (Step Over)”，而不是“单步进入”或“单步跳出”，然后单击它。接下来

.....



ActivitiesQt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

FileEditViewBuildDebugAnalyzeToolsWindowHelp

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

51 * prime. These numbers were chosen because the

52 * $2^{31} - kLargePrime - 1$.

53 */

54 static const int kLargePrime = 16908799;

55 static const int kSmallPrime = 127;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the fir

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower cas

64 * lower-case letters are always less than

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kL

68 }

69 return hashVal;

70 }

71 }

DebuggerGDB for "NameHash"

LevelFunction

1nameHash

2studentMain

3std::Function_handler<int (), Q

4GThreadStd::run()

5??

6start thread

NameValue

__for_begin@0x7fffc6058c78std::string::iterator

__for_end@0x7fffc6058c80std::string::iterator

__for_range"AdaLovelace"std::string &&

ch'a'970x61char

first"Ada"std::string

hashVal0int

kLargePrime16908799int

kSmallPrime127int

last"Lovelace"std::string

Type

std::string::iterator

std::string::iterator

std::string &&

char

std::string

int

int

int

std::string

NameValue

Type

Views

Threads

Call

Type to locate (Ctrl...

1Issues

2Search Results

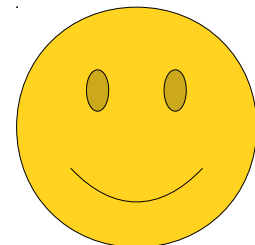
3Application Output

4Compile Output

5QML Debugger Console

7Version Control

8Test Results



……你的窗口应该会变成类似这样。

ActivitiesQt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

FileEditViewBuildDebugAnalyzeToolsWindowHelp

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

51 * prime. These numbers were chosen because the

52 * $2^{31} - kLargePrime - 1$.

53 */

54 static const int kLargePrime = 16908799;

55 static const int kSmallPrime = 127;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the first

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case

64 * lower-case letters are always less than

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69 return hashVal;

70 }

71 }

DebuggerGDB for "NameHash"

LevelFunction

1nameHash

2studentMain

3std::Function_handler<int (), QThreadStd::run()

4GThreadStd::run()

5??

6start thread

NameValue

Type

for_begin@0x7fffc6058c78std::string::iterator

for_end@0x7fffc6058c80std::string::iterator

for_range"AdaLovelace"std::string &&

ch'a'970x61char

first"Ada"std::string

hashVal0int

kLargePrime16908799int

kSmallPrime127int

last"Lovelace"std::string

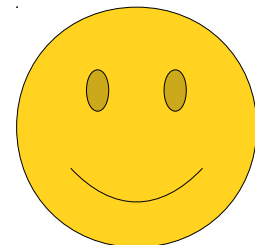
Views

Threads

all)

Type to locate (Ctrl+Shift+F)

1Issues2Search Results3Application Output4Compile Output5QML Debugger Console7Version Control8Test Results



好了！有几处发生了变化。我们来看看。

ActivitiesQt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

FileEditViewBuildDebugAnalyzeToolsWindowHelp

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

51 * prime. These numbers were chosen because the

52 * $2^{31} - kLargePrime - 1$.

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66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kL

68 }

69 return hashVal;

70 }

71 }

DebuggerGDB for "NameHash"

LevelFunction

1nameHash

2studentMain

3std::_Function_handler<int (), G

4GThreadStd::run()

5??

6start thread

NameValue

Value

Type

for_begin@0x7fffc6058c78std::string::iterator

for_end@0x7fffc6058c80std::string::iterator

for_range"AdaLovelace"std::string &&

ch'a'970x61char

first"Ada"std::string

hashVal0int

kLargePrime16908799int

kSmallPrime127int

last"Lovelace"std::string

Views

Threads

Call

Type to locate (Ctrl...

1Issues

2Search Results

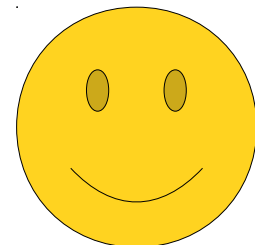
3Application Output

4Compile Output

5QML Debugger Console

7Version Control

8Test Results



首先，黄色箭头现在指向第 67 行。

ActivitiesQt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

FileEditViewBuildDebugAnalyzeToolsWindowHelp

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

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61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower cas

64 * lower-case letters are always less than

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kL

68 }

69 return hashVal;

70 }

71 }

Name

Value

Type

for_begin @0x7ffc6058c78 std::string::iterator

for_end @0x7ffc6058c80 std::string::iterator

for_range "AdaLovelace" std::string &&

ch 'a' 97 0x61 char

first "Ada" std::string

hashVal 0 int

kLargePrime 16908799 int

kSmallPrime 127 int

last "Lovelace" std::string

DebuggerGDB for "NameHash"

LevelFunction

1nameHash

2studentMain

3std::_Function_handler<int (), G

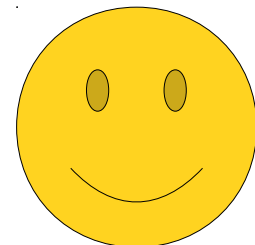
4GThreadStd::run()

5??

6start thread

Type to locate (Ctrl...

1Issues2Search Results3Application Output4Compile Output5QML Debugger Console7Version Control8Test Results



我们已经来到刚才暂停位置的下一行。你刚刚只执行了一行代码！

ActivitiesQt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

FileEditViewBuildDebugAnalyzeToolsWindowHelp

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

51 * prime. These numbers were chosen because the

52 * $2^{31} - kLargePrime - 1$.

53 */

54 static const int kLargePrime = 16908799;

55 static const int kSmallPrime = 127;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the first

60 * name, updating the hash at each step.

61 */

62 for (char ch: first + last) {

63 /* Convert the input character to lower case

64 * lower-case letters are always less than

65 */

66 ch = tolower(ch);

67 hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;

68 }

69 return hashVal;

70 }

71 }

DebuggerGDB for "NameHash"

LevelFunction

1nameHash

2studentMain

3std::Function_handler<int (), QThreadStd::run()

4GThreadStd::run()

5??

6start thread

NameValue

__for_begin@0x7fffc6058c78std::string::iterator

__for_end@0x7fffc6058c80std::string::iterator

__for_range"AdaLovelace"std::string &&

ch'a'970x61char

first"Ada"std::string

hashVal0int

kLargePrime16908799int

kSmallPrime127int

last"Lovelace"std::string

Type

std::string::iterator

std::string::iterator

std::string &&

char

std::string

int

int

int

std::string

NameValue

Type

那么，刚才那行代码做了什么？

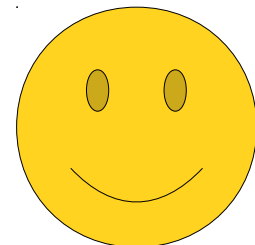
Views

Threads

Call

Type to locate (Ctrl+K)

1Issues2Search Results3Application Output4Compile Output5QML Debugger Console7Version Control8Test Results



Activities

Qt Creator

Jan 4 3:42 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

248

249

250

251

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266

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270

271

*/

int nameHash(string first, string last){

/* This hashing scheme needs two prime numbers.

* prime. These numbers were chosen because the

* $2^{31} - kLargePrime - 1$.

*/

static const int kLargePrime = 16908799;

static const int kSmallPrime = 127;

int hashVal = 0;

/* Iterate across all the characters in the fir

* name, updating the hash at each step.

*/

for (char ch: first + last) {

/* Convert the input character to lower cas

* lower-case letters are always less than

*/

ch = tolower(ch);

hashVal = (kSmallPrime * hashVal + ch) % k

}

return hashVal;

}

Debugger

GDB for "NameHash"

Level Function

1 nameHash

2 studentMain

3 std::Function_handler<int (), G

4 GThreadStd::run()

5 ??

6 start thread

248

249

250

251

252

253

254

255

256

257

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259

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261

262

263

264

265

266

267

268

269

270

271

Name

Value

Type

__for_begin

@0x7fffc6058c78

std::string::iterator

__for_end

@0x7fffc6058c80

std::string::iterator

__for_range

"AdaLovelace"

std::string &&

ch

'a'

97

0x61

char

first

"Ada"

std::string

hashVal

0

int

kLargePrime

16908799

int

kSmallPrime

127

int

last

"Lovelace"

std::string

Type to locate (Ctrl...

1 Issues

2 Search Results

3 Application Output

4 Compile Output

5 QML Debugger Console

7 Version Control

8 Test Results



这行代码把 ch 转换成小写。tolower 函数接收一个字符并返回其小写形式，因此这里会用 ch 自身的小写形式覆盖 ch。

查看右侧的变量值面板，就能直接看到这个变化！



```
50 int nameHash(string first, string last){
51     /* This hashing scheme needs two prime numbers:
52      * prime. These numbers were chosen because the
53      *  $2^{31} - kLargePrime - 1$ .
54      */
55     static const int kLargePrime = 16908799;
56     static const int kSmallPrime = 127;
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71 }
```

	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'a' 97	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Name	Value	Type
------	-------	------

Debugger - GDB for "NameHash" - Threads: #12 NameHash - Stopped: "end-stepping-range".										
Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	g) ...eHash.cpp	66	...	55555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595						
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc						
4	GThreadStd::run()			0x5555555f9476						
5	??			0x7ffff6143d84						
6	start_thread	pthread_create.c	463	0x7ffff6257590						

注意，ch 的值已经从 A 变成 a，也就是转换成了小写。



```
50 int nameHash(string first, string last){
51     /* This hashing scheme needs two prime numbers;
52     * prime. These numbers were chosen because the
53     *  $2^{31} - kLargePrime - 1$ .
54     */
55     static const int kLargePrime = 16908799;
56     static const int kSmallPrime = 127;
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60     * name, updating the hash at each step.
61     */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64         * lower-case letters are always less than
65         */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71 }
```

	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"Ada Lovelace"	std::string &&
ch	'a' 97	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Name	Value	Type
------	-------	------

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	g) ...eHash.cpp	66	...	...		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

可以看到，这个值显示为红色，而其他值仍是黑色。



```
50 int nameHash(string first, string last){
51     /* This hashing scheme needs two prime numbers;
52        * prime. These numbers were chosen because the
53        *  $2^{31} - kLargePrime - 1$ .
54        */
55     static const int kLargePrime = 16908799;
56     static const int kSmallPrime = 127;
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60        * name, updating the hash at each step.
61        */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64            * lower-case letters are always less than
65            */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71 }
```

	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"Ada Lovelace"	std::string &&
ch	'a'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Name	Value	Type
------	-------	------

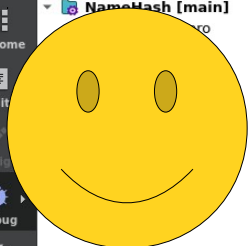
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	g) ...eHash.cpp	66	...	...		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

红色表示该值自上一步之后发生了变化。这是一种非常方便的方式，可以在程序运行过程中追踪哪些内容正在改变。



```
50 int nameHash(string first, string last){
51     /* This hashing scheme needs two prime numbers;
52     * prime. These numbers were chosen because the
53     *  $2^{31} - kLargePrime - 1$ .
54     */
55     static const int kLargePrime = 16908799;
56     static const int kSmallPrime = 127;
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60     * name, updating the hash at each step.
61     */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64         * lower-case letters are always less than
65         */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

	Value	Type
for_begin	@0x7ffc6058c78	std::string::iterator
for_end	@0x7ffc6058c80	std::string::iterator
for_range	"Ada Lovelace"	std::string &&
ch	'a'	char
first	"Ada"	std::string
hashVal	0	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

Name	Value	Type
------	-------	------

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	g) ...eHash.cpp	66	...	...		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

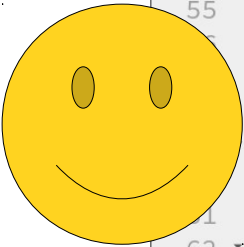
Activities Qt Creator Jan 4 3:42 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Now looking at line 67 of the program.



```
54
55
56 // ...
57 int hashVal = 0;
58
59 /* Iterate across all the characters in the first
60 * name, updating the hash at each step.
61 */
62 for (char ch: first + last) {
63     /* Convert the input character to lower case
64     * lower-case letters are always less than
65     */
66     ch = tolower(ch);
67     hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68 }
69 return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	67	0x555555b6790
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	...	...eHash.cpp	66	...	...		(all)

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:42 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Welcome Edit Design Debug Projects Help

54
55 start = first + last - 1; // first and last are pointers to the first and last characters of the string
56
57 int hashVal = 0;
58
59 /* Iterate across all the characters in the first and last characters of the string, updating the hash at each step.
60 */
61 for (char ch: first + last) {
62 /* Convert the input character to lower case. Lower-case letters are always less than
63 * upper-case letters, so we can use the same hash for both. */
64 ch = tolower(ch);
65
66 hashVal = (kSmallPrime * hashVal + ch) % kMaxHashVal;
67
68 }
69 return hashVal;
70 }
71

67 hashVal = (kSmallPrime * hashVal + ch) % kMaxHashVal;

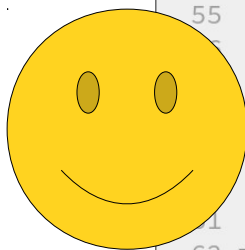
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	67	0x555555b6790
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()			0x555555b616c
4	GThreadStd::run()			0x555555b69476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

Number	Func	File	Line	Address	Condition	Ignore	Threads
1	...	...eHash.cpp	66	...	...		(all)

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

坦白说，这一行代码相当紧凑。它会对多个不同的值进行一段看起来有些奇怪的数学计算。



Activities Qt Creator Jan 4 3:42 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Welcome Edit Design Debug Projects Help

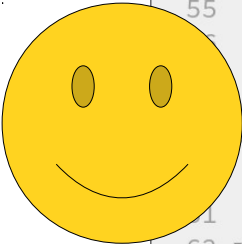
54
55 start = 127;
56
57 int hashVal = 0;
58
59 /* Iterate across all the characters in the first
60 * name, updating the hash at each step.
61 */
62 for (char ch: first + last) {
63 /* Convert the input character to lower case
64 * lower-case letters are always less than
65 */
66 ch = tolower(ch);
67 hashVal = (kSmallPrime * hashVal + ch) % kSmallPrime;
68 }
69 return hashVal;
70 }
71

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	...eHash.cpp	66	...5555b6782			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

但从根本上说，它只是根据这些值计算出一个结果，然后把结果保存到 hashVal 中。



Activities Qt Creator Jan 4 3:42 PM NameHash.cpp @ NameHash [main] - Qt Creator

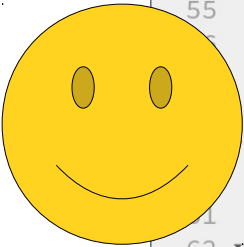
File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Welcome Edit Design Debug Projects Help

执行这一行，看看会发生什么！



```
54
55
56 start = 127;
57
58 int hashVal = 0;
59
60 /* Iterate across all the characters in the first
61 * name, updating the hash at each step.
62 */
63 for (char ch: first + last) {
64     /* Convert the input character to lower case
65     * lower-case letters are always less than
66     */
67     ch = tolower(ch);
68     hashVal = (kSmallPrime * hashVal + ch) % kMaxVal;
69 }
70 return hashVal;
71 }
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	67	0x555555b6790	1	...	...eHash.cpp	66	...			(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl+Shift+F) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities

Qt Creator

Jan 4 3:48 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

248

48

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69

70

71

*/

int nameHash(string first, string last){

/* This hashing scheme needs two prime numbers:

* prime. These numbers were chosen because the

* $2^{31} - kLargePrime - 1$.

*/

static const int kLargePrime = 16908799;

static const int kSmallPrime = 127;

int hashVal = 0;

*/

/* Iterate across all the characters in the fir

* name, updating the hash at each step.

*/

for (char ch: first + last) {

/* Convert the input character to lower cas

* lower-case letters are always less than

*/

ch = tolower(ch);

hashVal = (kSmallPrime * hashVal + ch) % kL

}

return hashVal;

}

Name

Value

Type

for_begin

@0x7fffc6058c78

std::string::iterator

for_end

@0x7fffc6058c80

std::string::iterator

for_range

"AdaLovelace"

std::string &&

ch

'a'

97

0x61

char

first

"Ada"

std::string

hashVal

97

int

kLargePrime

16908799

int

kSmallPrime

127

int

last

"Lovelace"

std::string

Debugger

GDB for "NameH

Level

Function

1

nameHash

2

studentMain

3

std::_Function_handler<int (), QtGui::startBackgroundDeve...

4

GThreadStd::run()

5

??

6

start thread

pthread create.c

463

0x7ffff6257590

Type to locate (Ctrl...

1 Issues

2 Search Results

3 Application Output

4 Compile Output

5 QML Debugger Console

7 Version Control

8 Test Results

.....你会看到类似这样的结果!



Activities

Qt Creator

Jan 4 3:48 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

NameHash [main]

NameHash.pro

Sources

NameHash.cpp

248

48

49

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51

52

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54

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56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

*/

int nameHash(string first, string last){

/* This hashing scheme needs two prime numbers:

* prime. These numbers were chosen because the

* $2^{31} - kLargePrime - 1$.

*/

static const int kLargePrime = 16908799;

static const int kSmallPrime = 127;

int hashVal = 0;

*/

Iterate across all the characters in the fir

* name, updating the hash at each step.

*/

for (char ch: first + last) {

/* Convert the input character to lower cas

* lower-case letters are always less than

*/

ch = tolower(ch);

hashVal = (kSmallPrime * hashVal + ch) % kL

}

return hashVal;

}

Name

Value

Type

for_begin

@0x7fffc6058c78

std::string::iterator

for_end

@0x7fffc6058c80

std::string::iterator

for_range

"AdaLovelace"

std::string &&

ch

'a'

97

0x61

char

first

"Ada"

std::string

hashVal

97

int

kLargePrime

16908799

int

kSmallPrime

127

int

last

"Lovelace"

std::string

Debugger

GDB for "NameH

Level

Function

1

nameHash

2

studentMain

3

std::_Function_handler<int (), QtGui::startBackgroundDeve...

4

GThreadStd::run()

5

??

6

start thread

pthread create.c

463

0x7ffff6257590

Type to locate (Ctrl...

1 Issues

2 Search Results

3 Application Output

4 Compile Output

5 QML Debugger Console

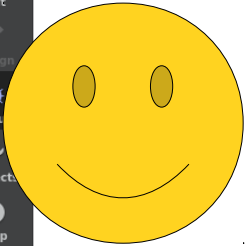
7 Version Control

8 Test Results

看看哪些内容发生了变化。



首先, hashVal 中保存的值变成了 97。它显示为红色, 因此我们知道它发生了变化; 其他内容都不是红色, 所以没有其他值改变。



```
51  /* This hashing scheme needs two prime numbers;
52  * prime. These numbers were chosen because the
53  *  $2^{31} - kLargePrime - 1$ .
54  */
55  static const int kLargePrime = 16908799;
56  static const int kSmallPrime = 127;
57
58  int hashVal = 0;
59
60  /* Iterate across all the characters in the first
61  * name, updating the hash at each step.
62  */
63  for (char ch: first + last) {
64      /* Convert the input character to lower case
65      * lower-case letters are always less than
66      */
67      ch = tolower(ch);
68      hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
69  }
70  return hashVal;
71 }
```

Value	Type
@0x7fffc6058c78	std::string::iterator
@0x7fffc6058c80	std::string::iterator
"AdaLovelace"	std::string &&
ch	char
hashVal	97
kLargePrime	16908799
kSmallPrime	127
last	"Lovelace"

Name	Value	Type
------	-------	------

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb	1	...	g) ...eHash.cpp	66	...	5555b6782		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Activities Qt Creator Jan 4 3:48 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51     * prime. These numbers were chosen because the
52     *  $2^{31} - kLargePrime$ 
53     */
54     static
55     static
56
57     int has
58
59     /* Iter
60     * name
61     */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64         * lower-case letters are always less than
65         */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kL
68     }
69     return hashVal;
70 }
71
```

Variable Inspector:

Name	Value	Type
__for_begin	@0x7fffc6058c78	std::string::iterator
__for_end	@0x7fffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'a' 97	char
first	"Ada"	std::string
hashVal	97	int
kLargePrime	16908799	int
kSmallPrime	127	int

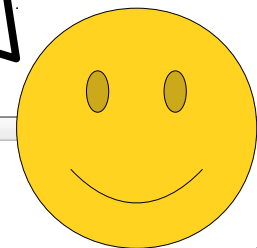
Debugger: GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb	1	...	g) ...eHash.cpp	66	...	5555b6782		(all)
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

其次，我们回到了 for 循环顶部，黄色箭头正指向那里。之所以回到这里，是因为它就是下一条将要执行的代码。



Activities Qt Creator Jan 4 3:48 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

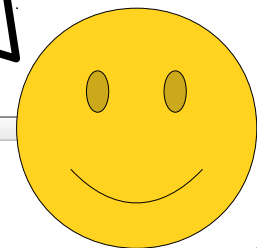
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters in the string.
60      * nameHash will iterate over the characters in the string.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case.
64          * lower-case letters are always less than 128.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

我们刚刚逐步执行完了这个循环的一次迭代！



Activities Qt Creator Jan 4 3:48 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

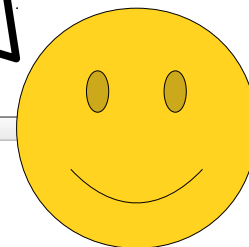
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters in the strings.
60      * nameHash will iterate over the characters in the strings.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case.
64          * lower-case letters are always less than 128.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::function_handler<int (), QtGui::startBackgroundEventLoop()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

再来一次!



Activities Qt Creator Jan 4 3:48 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

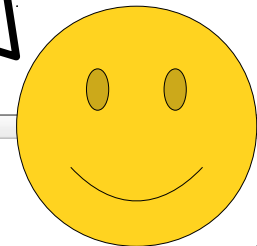
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters in the string.
60      * nameHash will be called with the first and last
61      * characters of the string.
62     */
63     for (char ch: first + last) {
64         /* Convert the input character to lower case.
65          * lower-case letters are always less than
66          * upper-case letters.
67          */
68         ch = tolower(ch);
69         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
70     }
71     return hashVal;
72 }
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::Function_handler<int (), QtGui::startBackgroundThread::run()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

再次把鼠标移到“单步跳过 (Step Over)”按钮上，确认提示文字正确，然后单击它。



Activities Qt Creator Jan 4 3:50 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

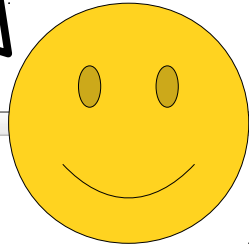
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over each character in the first and last names.
60      * nameHash will calculate the hash for each character.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case.
64          * lower-case letters are always less than 128.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()			0x5555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

现在到了这里！注意，ch 的值变成了 d，这是 Ada 的第二个字母。



Activities Qt Creator Jan 4 3:50 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

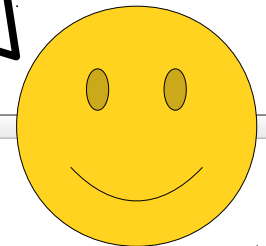
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters in the string.
60      * nameHash will return the hash value for the string.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case.
64          * lower-case letters are always less than 128.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped at breakpoint 1 in thread 12.

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	66	0x555555b6782
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundThreadEventLoop...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

再次单击“单步跳过 (Step Over)”，执行这一行代码。



Activities Qt Creator Jan 4 3:52 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

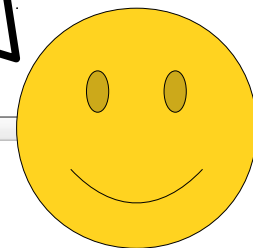
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters in the first and last strings.
60      * nameHash will calculate the hash value for the entire string.
61      */
62     for (char ch: first + last) {
63         /* Convert the input character to lower case.
64          * lower-case letters are always less than 128.
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	67	0x555555b6790
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

现在应该停在这里。注意，没有任何值发生变化。这很合理，因为我们只是把小写 d 转换成了小写 d。



Activities Qt Creator Jan 4 3:52 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

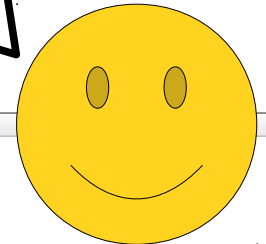
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate over the characters of the first and last names.
60      * nameHash will calculate the hash value for the first and last
61      * names.
62      */
63     for (char ch: first + last) {
64         /* Convert the input character to lower case.
65          * lower-case letters are always less than 128.
66          */
67         ch = tolower(ch);
68         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
69     }
70     return hashVal;
71 }
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	67	0x555555b6790
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

再单击一次“单步跳过 (Step Over)”。



Activities Qt Creator Jan 4 3:53 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     // ...
56
67
68 }
69 return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

看这里!

现在程序会停在这个位置。图中遮住了 hashVal 的值，因为此时你应该已经能够从右侧面板读取它。这是提交作业时需要填写的特殊值!



Activities Qt Creator Jan 4 3:53 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

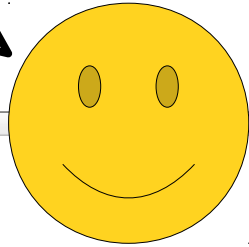
Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Name Value Type

__for_begin	@0x7ffff6058c78	std::string::iterator
__for_end	@0x7ffff6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'd' 100	0x64 char
first	"Ada"	std::string
hashVal	????	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

在结束调试器这一部分之前，我们还想介绍两个在调试程序时可能很有用的小技巧。



Activities Qt Creator Jan 4 3:53 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51     * prime. These numbers were chosen because the
52     *  $2^{31} - kLargePrime - 1$ .
53     */
54
55     for (char ch: first + last) {
56         /* Convert the input character to lower case
57         * lower-case letters are always less than
58         */
59         ch = tolower(ch);
60         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
61     }
62     return hashVal;
63 }
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
4	GThreadStd::run()			0x5555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

首先，单击之前在程序中设置的断点。单击后……



Activities Qt Creator Jan 4 3:57 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

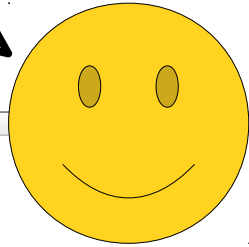
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - \text{kLargePrime} - 1$ .
53      */
54
61
62     for (char ch: first + last) {
63         /* Convert the input character to lower case
64          * lower-case letters are always less than
65          */
66         ch = tolower(ch);
67         hashVal = (kSmallPrime * hashVal + ch) % kLargePrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	nameHash	NameHash.cpp	62	0x555555b67cb
2	studentMain	NameHash.cpp	31	0x555555b6595
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc
4	GThreadStd::run()			0x555555f9476
5	??			0x7ffff6143d84
6	start_thread	pthread_create.c	463	0x7ffff6257590

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

……断点应该会被清除。此时若再次以调试模式运行程序，它就不会在这里暂停，因为已经没有断点要求它停下。



Activities Qt Creator Jan 4 3:57 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch : first)
63     {
64         /*
65          *
66          */
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

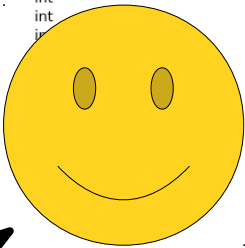
Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb								
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

现在再看看这些按钮。



Activities Qt Creator

Jan 4 3:57 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

48 */

49 int nameHash(string first, string last){

50 /* This hashing scheme needs two prime numbers;

51 * prime. These numbers were chosen because the

52 * $2^{31} - \text{kLargePrime} - 1$.

53 */

54 static const int kLargePrime = 16908799;

55 static const int kSmallPrime = 127;

56

57 int hashVal = 0;

58

59 /* Iterate across all the characters in the fir

60 * name, updating the hash at each step.

61 */

62 for (ch

63 /*

64 *

65 */

66 ch

67 has

68 }

69 return hashVal;

70 }

71

Name

Name	Value	Type
__for_begin	@0x7ffc6058c78	std::string::iterator
__for_end	@0x7ffc6058c80	std::string::iterator
__for_range	"AdaLovelace"	std::string &&
ch	'd' 100	char
first	"Ada"	std::string
hashVal	????	int
kLargePrime	16908799	int
kSmallPrime	127	int
last	"Lovelace"	std::string

把鼠标悬停在最右侧的按钮上。提示文字应为“单步跳出 (Step Out)”。

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb								
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEve...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 3:57 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

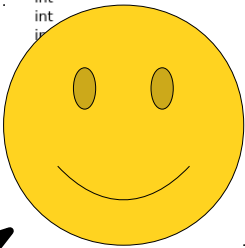
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because they are
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch : first)
63     {
64         /*
65          *
66          */
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Func	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb								
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackgroundEventLoop()>::operator()<int (), QtGui::startBackgroundEventLoop()>()			0x55555556161bc								
4	GThreadStd::run()			0x5555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

先不要单击。单击后，它会继续运行 nameHash 函数剩余的部分，直到该函数执行结束并返回。



Activities Qt Creator Jan 4 3:57 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

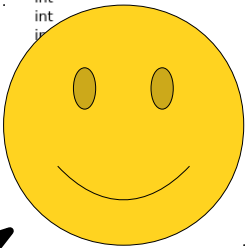
```
48 */
49 int nameHash(string first, string last){
50     /* This hashing scheme needs two prime numbers:
51      * prime. These numbers were chosen because the
52      *  $2^{31} - kLargePrime - 1$ .
53      */
54     static const int kLargePrime = 16908799;
55     static const int kSmallPrime = 127;
56
57     int hashVal = 0;
58
59     /* Iterate across all the characters in the first
60      * name, updating the hash at each step.
61      */
62     for (char ch : first)
63     {
64         /*
65          *
66          */
67         hashVal = (hashVal * kLargePrime + ch) % kSmallPrime;
68     }
69     return hashVal;
70 }
71
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address	Number	Funct	File	Line	Address	Condition	Ignore	Threads
1	nameHash	NameHash.cpp	62	0x555555b67cb								
2	studentMain	NameHash.cpp	31	0x555555b6595								
3	std::_Function_handler<int (), QtGui::startBackground...			0x5555556161bc								
4	GThreadStd::run()			0x555555f9476								
5	??			0x7ffff6143d84								
6	start_thread	pthread_create.c	463	0x7ffff6257590								

Type to locate (Ctrl+K) 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

现在单击这个按钮。如果前面的操作都正确……



Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVa
34     return 0;
35 }
36
37 /* This is
38 * to talk
39 * the mean
40 * of the i
41 *
42 * For thos
43 * treats e
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

returned value 1967457 int

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address	Number	Funcnt	File	Line	Address	Condition	Ignore	Threads
1	studentMain	NameHash.cpp	31	0x555555b6595								
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc								
3	GThreadStd::run()			0x5555555f9476								
4	??			0x7ffff6143d84								
5	start_thread	pthread_create.c	463	0x7ffff6257590								
6	clone	clone.S	95	0x7ffff5e30223								

Type to locate (Ctrl...

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVa
34     return 0;
35 }
36
37 /* This is
38 * to talk
39 * the mean
40 * of the i
41 *
42 * For thos
43 * treats e
```

Name Value Type

first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

returned value 1967457 int

先确认一下当前状态：我们现在究竟在哪里？

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address	Number	Funcnt	File	Line	Address	Condition	Ignore	Threads
1	studentMain	NameHash.cpp	31	0x555555b6595								
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc								
3	GThreadStd::run()			0x5555555f9476								
4	??			0x7ffff6143d84								
5	start_thread	pthread_create.c	463	0x7ffff6257590								
6	clone	clone.S	95	0x7ffff5e30223								

Type to locate (Ctrl... 1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Activities Qt Creator

Jan 4 4:02 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

20 #include "simpio.h" // for getLine

21 using namespace std;

22 /* Prototype for the nameHash function. This lets u

23 * in main and then define it later in the program.

24 */

25 int nameHash(string first, string last);

26

27 int main() {

28 string first = getLine("What is your first name

29 string last = getLine("What is your last name?

30

31 int hashValue = nameHash(first, last);

32

33 cout << "The hash of your name is: " << hashVa

34 return 0;

35 }

36

37 /* This is

38 * to talk

39 * the mean

40 * of the i

41 *

42 * For thos

43 * treats e

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	31	0x555555b6595
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
3	GThreadStd::run()			0x5555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



黄色箭头表明我们已经回到了 main 函数。很好！

Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVa
34     return 0;
35 }
36
37 /* This is th
38 * to talk m
39 * the meant
40 * of the in
41 *
42 * For those
43 * treats ea
```

Debugger GDB for "NameHash"

Name	Value	Type
first	"Ada"	std::string
hashValue	0	int
last	"Lovelace"	std::string

returned value 1967457 int

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



可以看到，nameHash 函数返回了 1967457。调试器帮我们看到了结果！

注意：在部分 Mac 上，这个数字可能不会显示。看不到也不用担心，照常继续即可。

Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
19 #include "simpio.h" // for getLine
20 using namespace std;
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVal
34     return 0;
35 }
36
37 /* This is the actual function that computes the ha
38 * to talk more about what hash functions do later
39 the meantime, think of it as a function that scr
40 the input and produces a number.
```

Name	Value	Type
hashValue	0	int

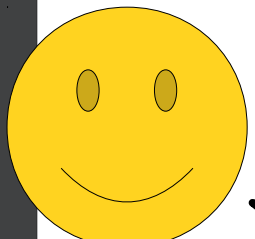
returned value 1967457 int

Name	Value	Type
------	-------	------

Views

Line	Address	Condition	Ignore	Threads
------	---------	-----------	--------	---------

Type to locate (Ctrl...



但查看这里可以发现，hashValue 并没有保存 1967457，尽管 nameHash 返回的正是这个值。

(你的系统上也可能显示一个非 0 的数值，这同样没问题。)



Activities Qt Creator

Jan 4 4:02 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

21 #include "simpio.h"

22 /* Prototype for nameHash function

23 * in main and

24 */

25 int nameHash(string first, string last)

26

27 int main() {

28 string first;

29 string last = getLine("What is your last name? ");

30

31 int hashValue = nameHash(first, last);

32

33 cout << "The hash of your name is: " << hashValue;

34 return 0;

35 }

36

37 /* This is the actual function that computes the hash

38 * to talk more about what hash functions do later

39 * the meantime, think of it as a function that scans

40 * of the input and produces a number.

41 *

42 * For those of you who are more mathematically inclined,

43 * treats each character in the input name as a number.

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	31	0x5555555b6595
2	std::Function_handler<int (&), QtGui::startBackgroundEventLoop>::operator()			0x55555556161bc
3	GThreadStd::run()			0x55555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

可是代码看起来明明是在把 nameHash 的返回值赋给 hashValue。究竟发生了什么？

returned value 1967457 int

Name	Value	Type
------	-------	------



Jan 4 4:02 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

19 #include "simpio.h"

20 using namespace std;

21

22 /* Prototype for nameHash() */

23 * in main and

24 */

25 int nameHash(string first, string last);

26

27 int main() {

28 string first = getLine("What is your first name? ");

29 string last = getLine("What is your last name? ");

30

31 int hashValue = nameHash(first, last);

32

33 cout << "The hash of your name is: " << hashValue << endl;

34 return 0;

35 }

36

37 /* This is the actual function that computes the hash value. It uses a simple algorithm to talk more about what hash functions do later. For the meantime, think of it as a function that scans the input string, computes the sum of the input and produces a number.

41 *

42 * For those of you who are more mathematically inclined, the function

43 * treats each character in the input name as a number, and computes the sum of the

44 * squares of the numbers. The result is then divided by the number of characters in the input name to produce the final hash value.

45 */

46 int nameHash(string first, string last) {

47 int sum = 0;

48 for (int i = 0; i < first.length(); i++)

49 sum += first[i] * first[i];

50 for (int i = 0; i < last.length(); i++)

51 sum += last[i] * last[i];

52 return sum / (first.length() + last.length());

53 }

Debugger: GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	31	0x5555555b6595
2	std::function_handler<int (), QtGui::startBackgroundEvent...			0x55555556161bc
3	GThreadStd::run()			0x55555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

21 #include "simpio.h"

22 using namespace std;

23 /* Prototype for nameHash function

24 * in main and

25 */

26 int nameHash(string first, string last);

27 int main() {

28 string first;

29 string last = getLine("What is your last name?");

30

31 int hashValue = nameHash(first, last);

32

33 cout << "The hash of your name is: " << hashValue;

34 return 0;

35 }

36

37 /* This is the actual function that computes the hash

38 * to talk more about what hash functions do later

39 * the meantime, think of it as a function that scans

40 * of the input and produces a number.

41 *

42 * For those of you who are more mathematically inclined,

43 * treats each character in the input name as a number.

此时 nameHash 已经返回了一个值；但因为我们正在逐步执行程序，所以这个返回值还没有真正赋给 hashValue。

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	31	0x555555b6595
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
3	GThreadStd::run()			0x5555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

Name	Value	Type
first	"Ada"	std::string
hashValue	1967457	int

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



Activities Qt Creator Jan 4 4:02 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

29 30 31 32 33 34 35 36 37 38 39 40 41 42 43

```
#include "simpio.h"
using namespace std;

/* Prototype for nameHash()
 * in main and
 */
int nameHash(string first, string last);

int main() {
    string first = getLine("What is your first name? ");
    string last = getLine("What is your last name? ");

    int hashValue = nameHash(first, last);

    cout << "The hash of your name is: " << hashValue << endl;
    return 0;
}

/* This is the actual function that computes the hash value.
 * to talk more about what hash functions do later
 * the meantime, think of it as a function that scans
 * of the input and produces a number.
 *
 * For those of you who are more mathematically inclined,
 * treats each character in the input name as a number and
 * computes the hash value based on that.
 */
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "function-finished".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	31	0x555555b6595
2	std::Function_handler<int (>, QtGui::startBackgroundThread)::_FUN			0x55555556161bc
3	GThreadStd::run()			0x5555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

使用一次“单步跳过 (Step Over)”，完成这一整行的执行。单击该按钮后，如果操作正确……

returned value 1967457 int

Name	Value	Type
------	-------	------



Activities Qt Creator

Jan 4 4:07 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

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43
44
45

```
/* Prototype for the nameHash function. This lets u  
* in main and then de... in the program.  
*/  
int nameHash(s...  
  
int main() {  
    string fir  
    string las  
  
    int hashValue = nameHash(first, last);  
  
    cout << "The hash of your name is: " << hashVal  
    return 0;  
}  
  
/* This is the actual function that computes the ha  
* to talk more about what hash functions do later  
* the meantime, think of it as a function that scr  
* of the input and produces a number.  
*  
* For those of you who are more mathematically inc  
* treats each character in the input name as a num  
* It then uses them as coefficients in a polynomial  
* F_p, where p is a large prime number, and evalu
```

.....正确的值会被保存进去（注意它显示为红色），同时程序移动到下一行。

Name	Value	Type
first	"Ada"	std::string
hashValue	1967457	int
last	"Lovelace"	std::string

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	33	0x555555b65b3
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
3	GThreadStd::run()			0x5555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffff5e30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results



Activities Qt Creator

Jan 4 4:07 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

Welcome Edit Design Debug Projects Help

```
21
22 /* Prototype for the nameHash function
23  * in main and
24  */
25 int nameHash(string first, string last) {
26
27 int main() {
28     string first = "Ada";
29     string last = "Lovelace";
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashValue;
34     return 0;
35 }
36
37 /* This is the actual function that computes the hash
38  * to talk more about what hash functions do later
39  * the meantime, think of it as a function that scans
40  * of the input and produces a number.
41  *
42  * For those of you who are more mathematically inclined,
43  * treats each character in the input name as a number
44  * It then uses them as coefficients in a polynomial
45  * F_p, where p is a large prime number, and evaluates
```

Debugger GDB for "NameHash" Threads: #12 NameHash Stopped: "end-stepping-range".

Level	Function	File	Line	Address
1	studentMain	NameHash.cpp	33	0x555555b65b3
2	std::Function_handler<int (), QtGui::startBackgroundEve...			0x55555556161bc
3	GThreadStd::run()			0x55555555f9476
4	??			0x7ffff6143d84
5	start_thread	pthread_create.c	463	0x7ffff6257590
6	clone	clone.S	95	0x7ffffe30223

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

到这里，我们已经看到了几乎所有需要关注的内容。无需继续一步步执行到结尾，直接让程序继续运行即可。

Activities Qt Creator Jan 4 4:07 PM NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects NameHash [main] NameHash.pro Sources NameHash.cpp

```
21
22 /* Prototype for the nameHash function. This lets u
23 * in main and then define it later in the program.
24 */
25 int nameHash(string first, string last);
26
27 int main() {
28     string first = getLine("What is your first name
29     string last = getLine("What is your last name?
30
31     int hashValue = nameHash(first, last);
32
33     cout << "The hash of your name is: " << hashVal
34     return 0;
35 }
36
37 /* This is the actual function that computes the h
38 * to talk more about what hash functions do later
39 * the meantime, think of it as a function that scr
40 * of the input and prod
```

Variable View:

Name	Value	Type
first	"Ada"	std::string
hashValue	1967457	int
last	"Lovelace"	std::string

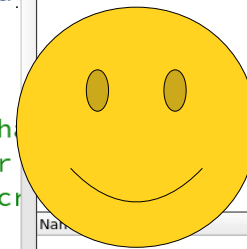
Stopped: "end-stepping-range".

Views: Funct File Line Address Condition Ignore Threads

1 Issues 2 Search Results 3 Application Output 4 Compile Output 5 QML Debugger Console 7 Version Control 8 Test Results

Type to locate (Ctrl...

单击这个按钮。鼠标悬停时会显示“继续 (Continue)” 。它表示：“解除暂停，并从当前位置继续运行程序。”



Activities NameHash Jan 4 4:08 PM

NameHash.cpp @ NameHash [main] - Qt Creator

File Edit View Build Debug Analyze Tools Window Help

Projects

- NameHash [main]
 - NameHash.pro
 - Sources
 - NameHash.cpp

21

22 `/* Prototype for the nameHash function. This lets u`

23 `* in main and then define it later in the program.`

24 `*/`

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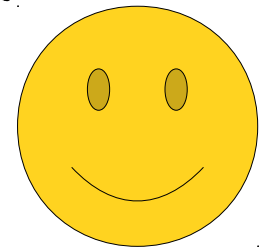
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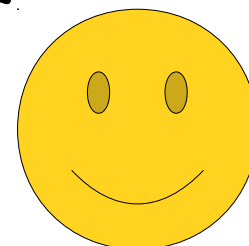
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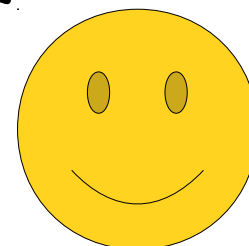
就是这样！现在你已经更加熟悉调试器了。



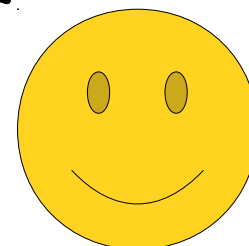
你已经知道如何设置断点，让程序在指定位置暂停。



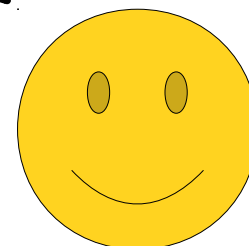
你已经知道如何读取调用栈，并查看局部变量的值。



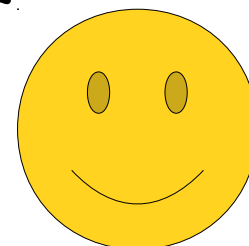
你已经知道如何逐步执行程序，并观察哪些值发生了变化。



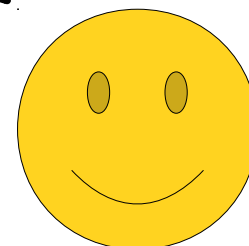
你已经知道如何让一个函数运行到结束，以及如何让整个程序
继续运行。



随着本季度编写的程序越来越复杂，你会越来越熟悉调试器，也会更清楚地看到程序的运行过程。



如果以后继续构建规模更大的软件，你会发现：熟练使用调试器
是一项非常有价值的技能！



希望这份教程对你有帮助。欢迎来到 CS106B!

